

Assignment for Test-Train Splitting & Polynomial Regression

Dataset: alloy-confp-train-data.csv

Target property: HV

Features: Al, Co, Cr, Cu, Fe, Ni

Goal: split the data into training and test sets, fit a linear regression model, evaluate RMSE and R2 on the test set, and then repeat the procedure using degree-3 polynomial features.

Linear Regression Results

Train-test split: test_size=0.2, shuffle=True, random_state=42

Model: LinearRegression()

Test RMSE: 71.96

Test R2: 0.83

Interpretation: the linear model fits the test data well and is the stronger model on this dataset compared with the polynomial version.

Polynomial Regression Results

PolynomialFeatures degree=3, include_bias=False

Model: LinearRegression() trained on the polynomial feature matrix

Test RMSE: 295.11

Test R2: -1.85

Conclusion: degree-3 polynomial regression performs worse here, so the simpler linear model is the better choice for this dataset.